22AI304 / 22AM304 / 22CD304 / 22CS304 / 22CT304 / 22IS304 / 22IT304 - PRINCIPLES OF PROGRAMMING LANGUAGES

PT1 - 2026 - 2027 - S3

Q. No.	**Questions**
UNIT 1	**SECTION 1**
1	A global cloud service provider is developing a real-time online banking platform with the following requirements: • Millions of customers should access the system simultaneously. • Customer account information must remain secure. • Financial transactions should execute with minimum response time. • Monthly reports should be generated automatically. • The software should be easy to maintain and support future expansion. The software team proposes two alternative system designs. Design A • Procedural Programming • Sequential execution • Manual report generation • Minimal modularity • Global variables Design B • Object-Oriented Programming • Concurrent Programming • Database Programming • Scripting Programming • Modular architecture with reusable components Recommend the most suitable design and justify your answer using the concepts of Object-Oriented Programming, Concurrent Programming, Database Programming, and Scripting Programming. (5 Marks)
2	A software development company is evaluating three coding styles for a large banking application. Style A count int; Style B int count; Style C int 1count; The project requirements are: • Follow standard programming language syntax. • Allow successful compilation. • Improve readability. • Minimize compiler errors. Identify the most appropriate coding style. and justify your answer usign the concepts of Syntax Rules,

	Identifier Rules and Compiler Error Detection. (5 Marks)
3	A software company is developing a voice-controlled virtual assistant for customer support. During testing, a customer says: "Book my ticket for tomorrow morning." The speech recognition system understands the sentence, but the compiler cannot execute it directly. Therefore, the system converts the request into a programming language statement before execution. Analyze the given scenario by distinguishing between natural language and artificial language. Justify why the programming language is required to execute the customer's request. (5 Marks)
4	(i) A multinational software company is developing a Programming Language Learning Platform. The system must satisfy the following requirements: • Programs should be easy for beginners to read. • Syntax errors should be detected immediately during compilation. • Programs should execute efficiently. • Coding standards should remain consistent across thousands of developers. The software architects propose two alternatives. Design A • Flexible syntax • Optional delimiters • Minimal syntax checking • Faster parser implementation Design B • Strict syntax rules • Mandatory delimiters • Early compiler error detection • Standardized program structure Recommend the most suitable design and justify your recommendation using the concepts of Syntax Rules, Error Detection, Readability, and Standardization. (3 Marks)
	(ii) Identify the conflicting design requirements that must be balanced while designing the language. (2 Marks)
UNIT 1	**SECTION 2**
5	A fintech company is developing an online banking system. During code review, the software engineer writes the following statement to deduct money from a customer's account: balance = balance - withdrawalAmount; The compiler first verifies that the statement follows the grammar of the programming language. It then performs additional checks to ensure that both variables have been declared, their data types are compatible, and the subtraction operation can be executed correctly before storing the updated balance. Only after these checks does the compiler generate executable code. Translate the above scenario into the logical sequence of semantic operations performed by the compiler before program execution. (5 Marks)

6	A company is developing software for controlling an autonomous railway signaling system. Since software failures may lead to accidents, engineers must choose one of the following approaches: Option A: Detect type errors during compilation. Option B: Detect type errors only while the train control software is running. The project manager requests a recommendation that maximizes reliability and minimizes runtime failures. Recommend the appropriate type checking approach with suitable justification. (5 Marks)
7	A software company is testing a payroll application. The following statements pass syntax checking, but one of them fails during semantic analysis. int salary = 50000; float bonus = 5000.5; salary = salary + bonus; The development team wants to understand why the compiler reports a semantic warning instead of a syntax error. Determine the semantic issue in the given program and recommend a suitable correction. Justify your answer. (5 Marks)
8	(i) A semantic analyzer reports the following message: Semantic Error: Undeclared Variable: totalMarks The syntax analyzer has already accepted the program. Analyze the above compiler output and reconstruct the most likely programming mistake that caused the error. (3 Marks)

	(ii) An inventory management application contains the following code:
	{
	  int stock = 50;
	}
	printf("%d", stock);
	The software engineer claims that the program is logically correct because the variable was declared before it was used.
	Critique the above statement by identifying the semantic error and explain why the compiler reports it.
	(2 Marks)

UNIT 2	**SECTION 1**
9	A software developer receives the following Python program. empName = "Ravi" empAge = "25" salary = 35000 empAge = empAge + 5 print(empName, empAge, salary) Analyze the program by identifying the conceptual errors and state the appropriate corrections for each error. (5 Marks)
10	A software developer writes the following program. salary = 25000

```
if salary > 30000:
    bonus = 5000
print(bonus)
```

Analyze the program and identify the problem caused by variable initialization. Demonstrate the error occurs and propose a suitable correction.

(5 Marks)

11 A courier company records the weights of packages collected during a day. The following program segment is used to process the data:

```
total_weight = 0.0
for package in range(4):
    weight = float(input("Enter package weight: "))
    total_weight += weight
average_weight = total_weight / 4
```

During a system review, the supervisor tests the program with the following package weights:

2.5 kg, 4.0 kg, 3.5 kg and 6.0 kg

Examine the given program and test data for the courier company. Determine the purpose, initial value and data type of total_weight, and calculate the final total_weight and average_weight after processing the four package weights.

(5 Marks)

12 (i) A university is redesigning its student database application. The software architect requests meaningful variable names that comply with Python coding standards.

Identify the information represented by the variables in the given program and recommend suitable domain-specific variable names that would make the billing process clearer and easier to maintain.

```
a="Arun"
b=91
c=95
d=(b+c)/2
print(a,d)
```

(3 Marks)

(ii) Evaluate the revised program and state two improvements in software quality resulting from the use of meaningful variable names.

(2 Marks)

UNIT 2 **SECTION 2**

13 A software company is developing a library management system. One developer stores the book price as a string, while another stores it as a float. Both implementations produce the same displayed output.

Evaluate which implementation is more appropriate for long-term software maintenance and justify your decision

(5 Marks)

14 A banking application receives the account balance as "25000.75" from user input. The programmer attempts to calculate annual interest directly using this value, resulting in an error.

Apply an appropriate type conversion technique to solve the problem and explain why the conversion is necessary before performing arithmetic operations.

(5 Marks)

15 A hospital management system is developed using Python. The programmer assigns the variable patient_id = 101 initially and later changes it to "P101" without modifying the program logic. This leads to

	inconsistent behavior in different modules. Apply your knowledge of type systems to identify the effect of dynamic typing during reassignment and suggest one suitable practice to avoid such issues. (5 Marks)
16	(i) An e-commerce application stores product details in a list, customer details in a dictionary, and tax rate in a float. During maintenance, a programmer accidentally modifies the tax rate and product collection without understanding mutability. Analyze the given data types, classify them as mutable or immutable, and state their impact on program behavior. (3 Marks)
	(ii) Recommend two best practices to avoid errors related to mutable and immutable data types. (2 Marks)
UNIT 3	**SECTION 1**
17	A smart home controller updates the room temperature every minute using the statement: temperature += 3. A beginner programmer rewrites it as: temperature = temperature + 3, and claims that compound assignment operators are only shorthand notation with no practical significance. Assess the beginner programmer's claim about the use of the compound assignment operator += in Python. Justify your assessment. Then, determine the final value of temperature for the following code segment by showing each stage of execution. temperature = 24 temperature += 3 temperature *= 2 (5 Marks)
18	During software testing, developers observe that the application crashes only when unexpected user inputs are entered. The program works correctly for valid inputs, but the testing team reports that users frequently make typing mistakes. Analyze the probable cause of the crashes. Recommend suitable improvements to the program design so that it behaves reliably even when users enter unexpected values. (5 Marks)
19	A hospital laboratory has developed a Python program to generate patient risk scores automatically. During software testing, auditors observe that identical patient data sometimes produces different risk classifications. On investigation, they find that programmers have written long expressions by mixing arithmetic, relational and logical operators without using parentheses. They also notice that some programmers have used is instead of == while comparing values. Before deploying the software, the development team asks you to review the program and recommend improvements to ensure reliable and maintainable code. Analyze the possible reasons for the inconsistent patient risk classifications by relating Python's operator precedence, associativity, and comparison operators. Recommend suitable programming practices to improve the correctness, readability, and maintainability of the expressions, and justify your recommendations. (5 Marks)
20	(i) A Python program developed five years ago is producing the correct output. The maintenance team is asked to add new features. However, the developers report that the program is difficult to understand because it uses poor variable names, inconsistent formatting and no logical organization. The management suggests leaving the program unchanged since it is already working correctly. Assess the management's decision with reference to the given scenario and justify your answer.

	(3 Marks)
	(ii) Recommend two modifications that should be made before extending the program. Justify how each modification supports future maintenance. (2 Marks)